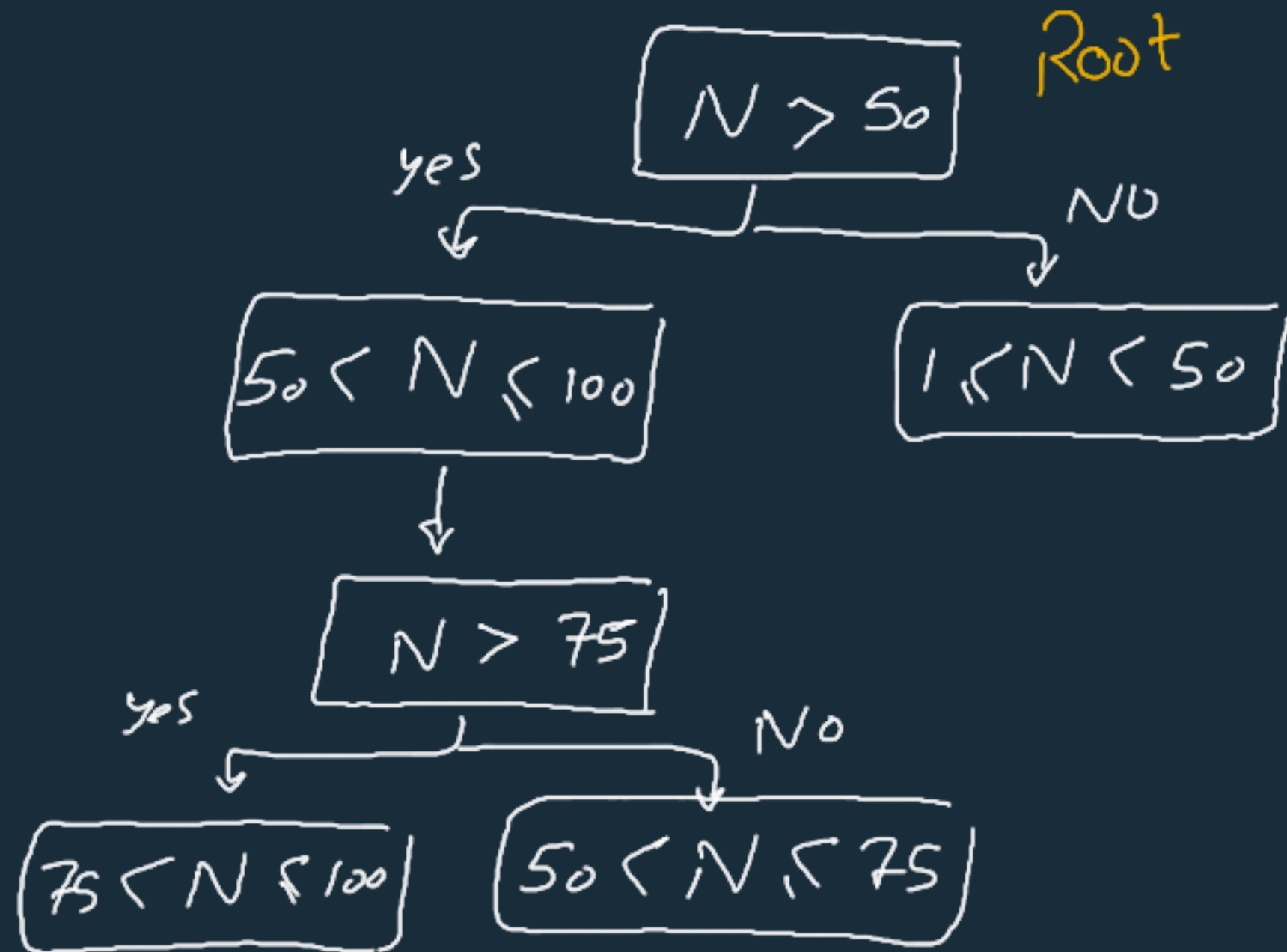


Decision Tree classifier

Guessing game (88) Target



* Questions

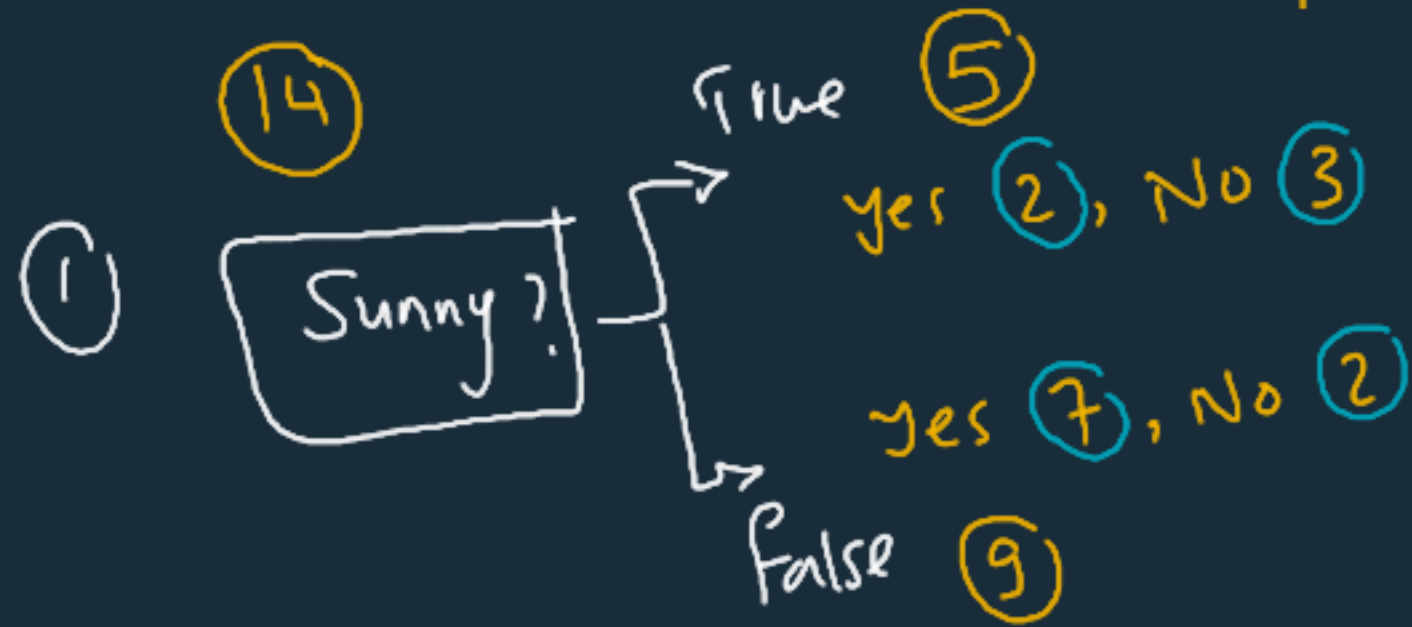
* Tree

* Depth

leaves
root

Impurity

Information Gain



$$Gini = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

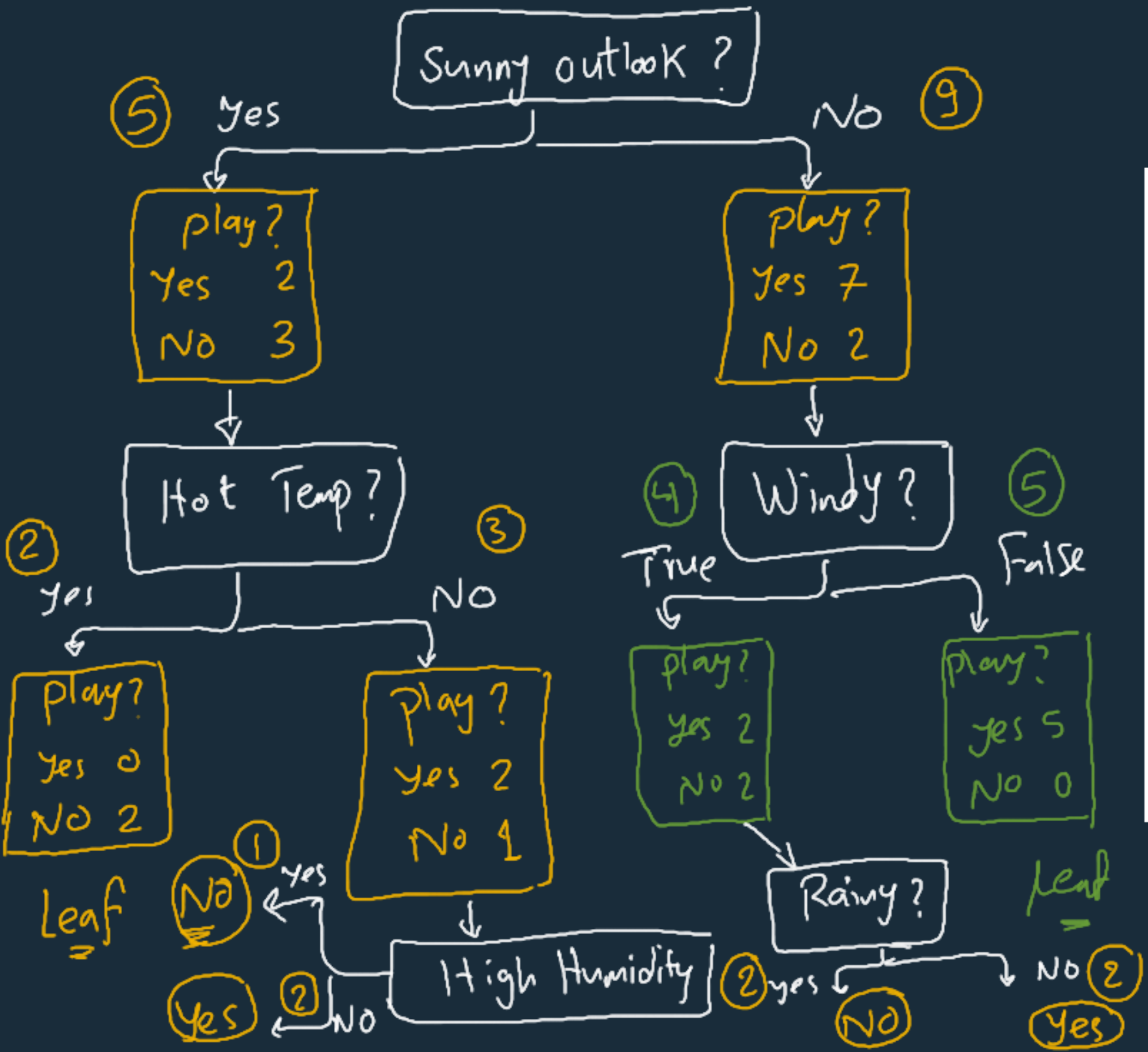
$$Gini = 1 - \left(\frac{7}{9}\right)^2 - \left(\frac{2}{9}\right)^2 = 0.35$$

$$\begin{aligned} \text{Total } Gini &= 0.48 \times \frac{5}{14} + 0.35 \times \frac{9}{14} \\ &= \underline{\underline{0.4}} \end{aligned}$$

Root

Features

Target

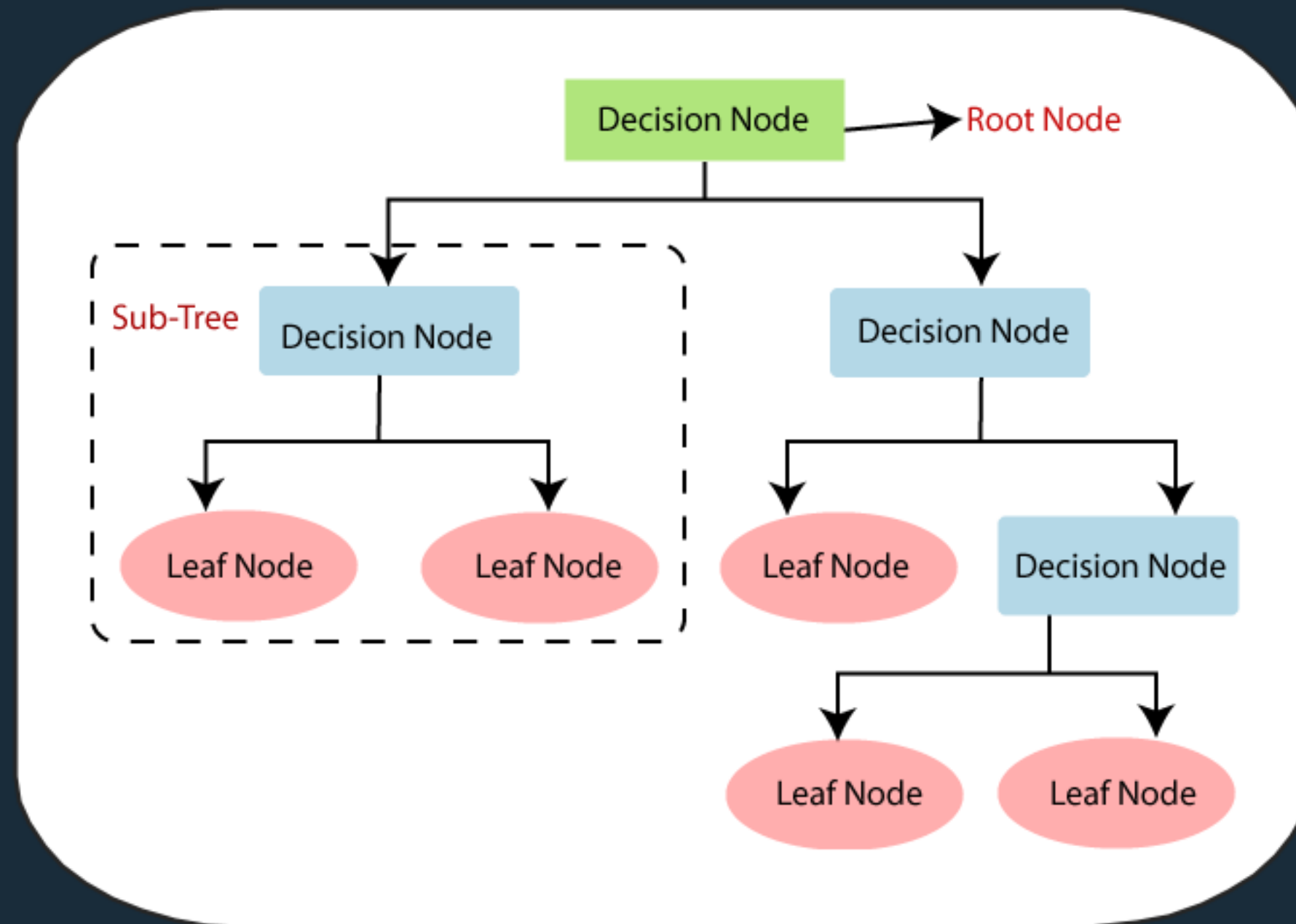


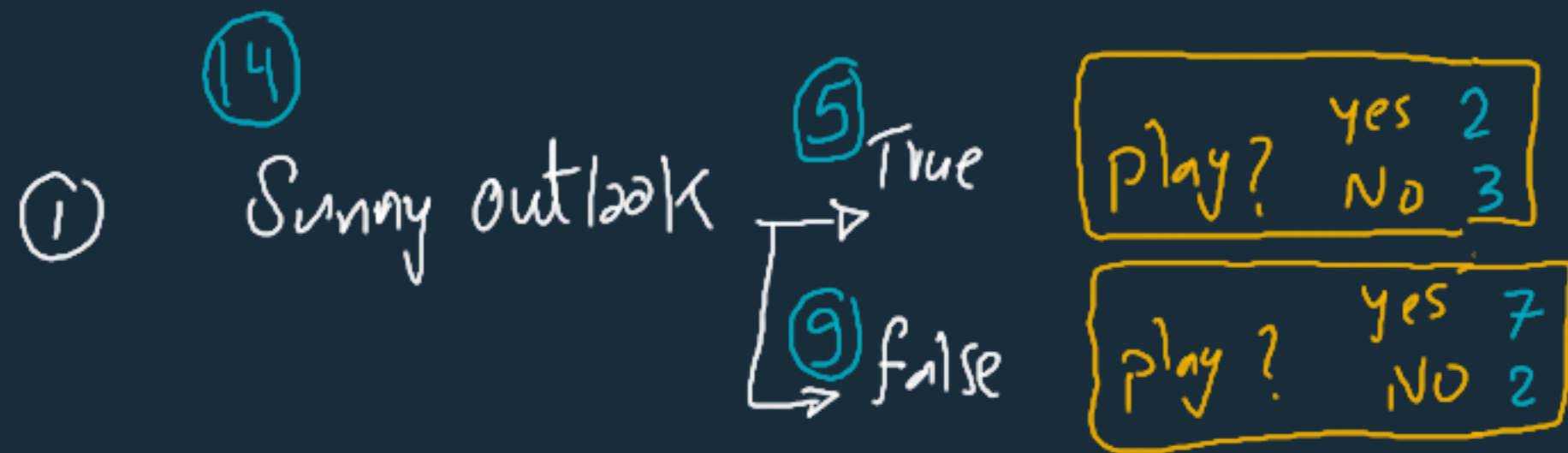
Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Depth = 3

Yes 9
No 5

Decision Tree





Impurity

Information Gain

Gini Index

↳ a measure of impurity

$$G_{\text{left}} = 1 - \left(\frac{7}{9}\right)^2 - \left(\frac{2}{9}\right)^2 = 0.35$$

left branch

$$G_{\text{right}} = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2 = 0.48$$

$$\text{Total Gini} = 0.35 \times \frac{9}{14} + 0.48 \times \frac{5}{14} = 0.41$$

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

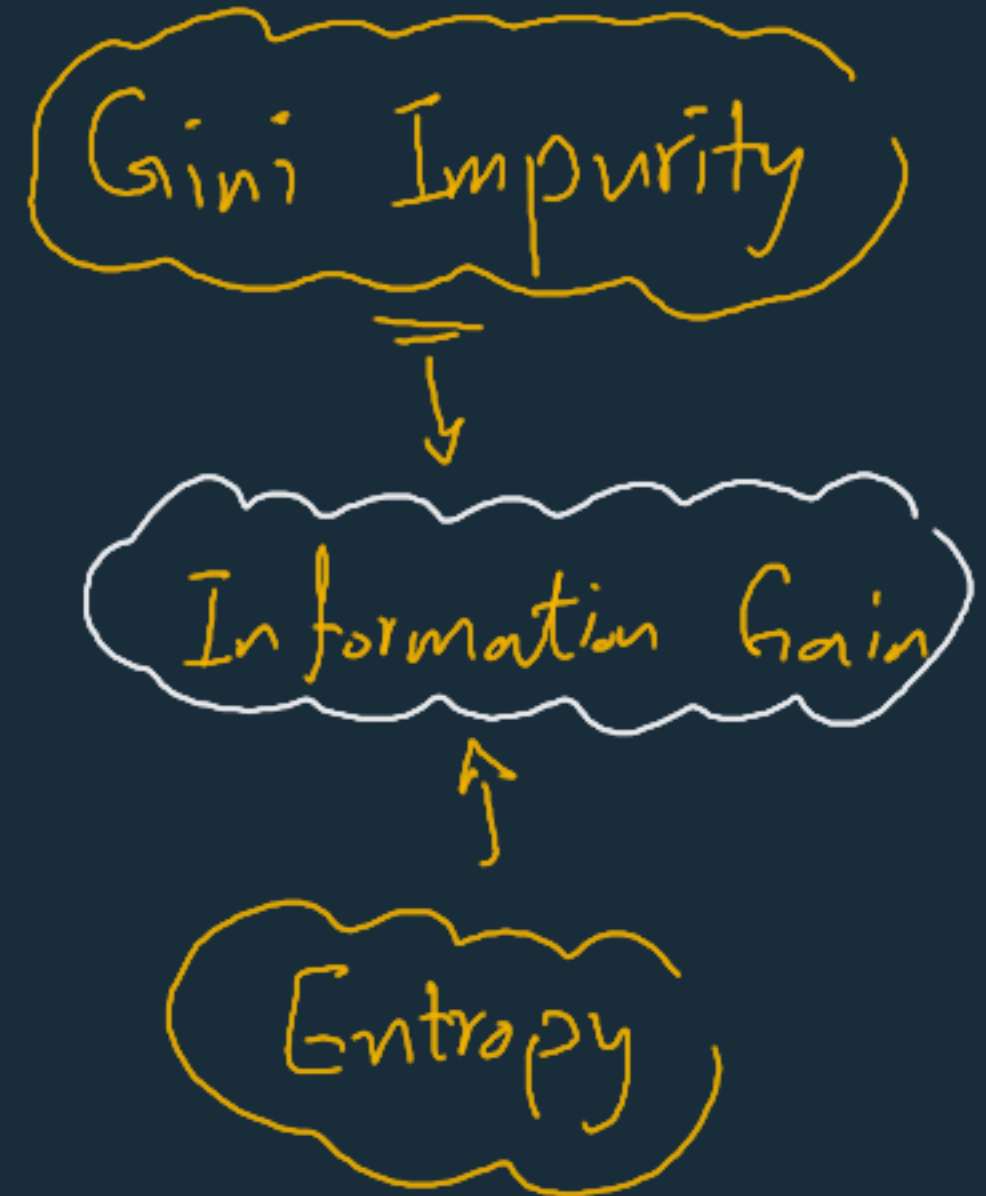
$$\bar{J} = \frac{m_{\text{left}}}{m} G_{\text{left}} + \frac{m_{\text{right}}}{m} G_{\text{right}}$$

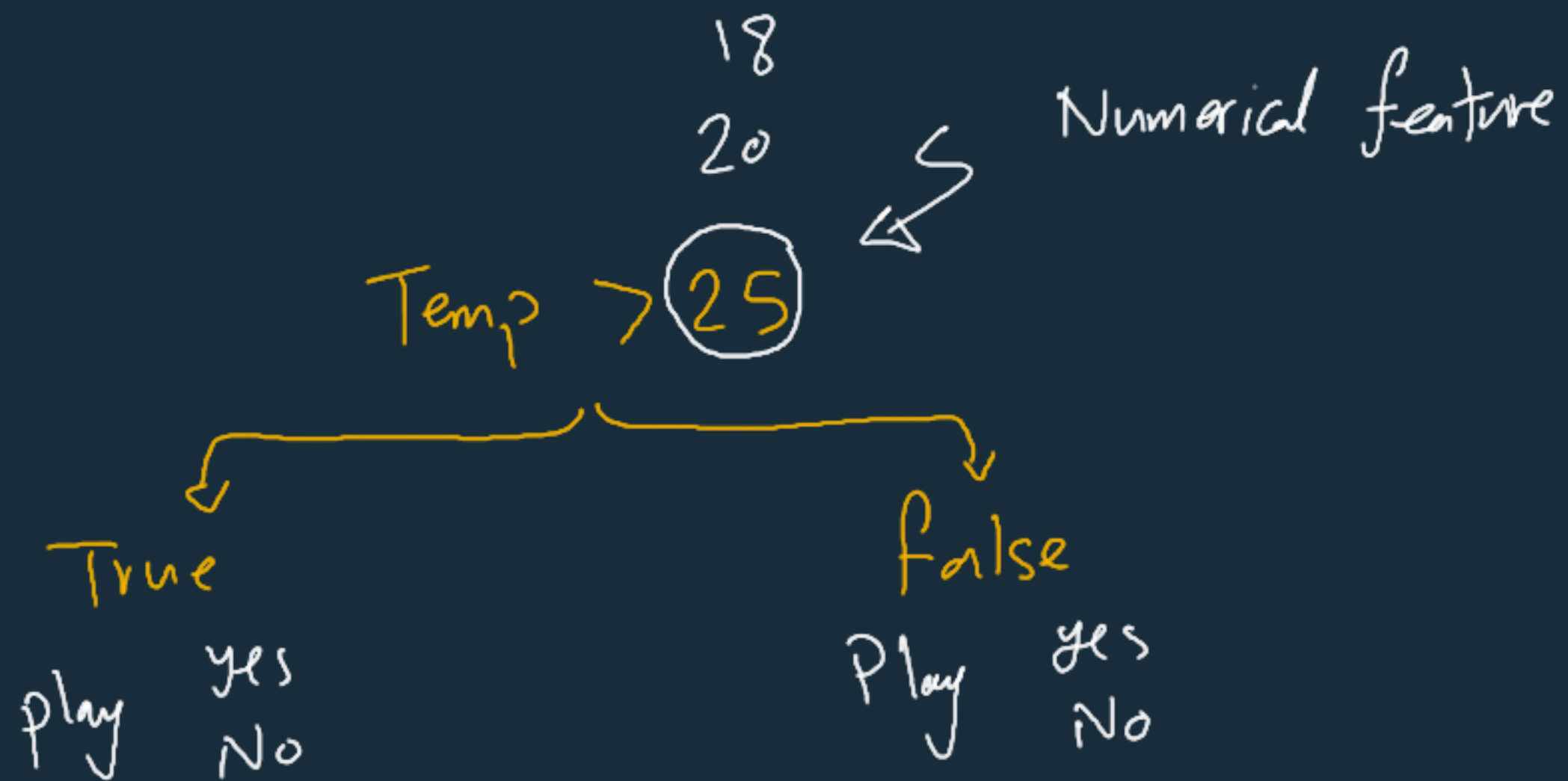
Cost Function

$$G = 1 - \sum_{k=1}^n p_k^2$$

→ Min Cost function

$$H = - \sum_{k=1}^n p \log_2(p_k)$$





Numerical Feature

① order it in ascending

② Calculate average for every two

③ Test for every result

Temp
 13
 14 {
 15 {
 15.5 {
 16 {
 17 {
 18 {
 18.5 {
 19 {
 20.5 {
 22 {

Temp < 14
 Temp < 15.5
 Gini Index
 Min Cost

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	35 Hot	High	False	No
Sunny	32 Hot	High	True	No
Overcast	33 Hot	High	False	Yes
Rainy	25 Mild	High	False	Yes
Rainy	15 Cool ←	Normal	False	Yes
Rainy	13 Cool ←	Normal	True	No
Overcast	16 Cool ←	Normal	True	Yes
Sunny	27 Mild	High	False	No
Sunny	18 Cool ←	Normal	False	Yes
Rainy	19 Mild ←	Normal	False	Yes
Sunny	22 Mild	Normal	True	Yes
Overcast	24 Mild	High	True	Yes
Overcast	30 Hot	Normal	False	Yes
Rainy	26 Mild	High	True	No

Q1 Temp ≤ 14

①

True

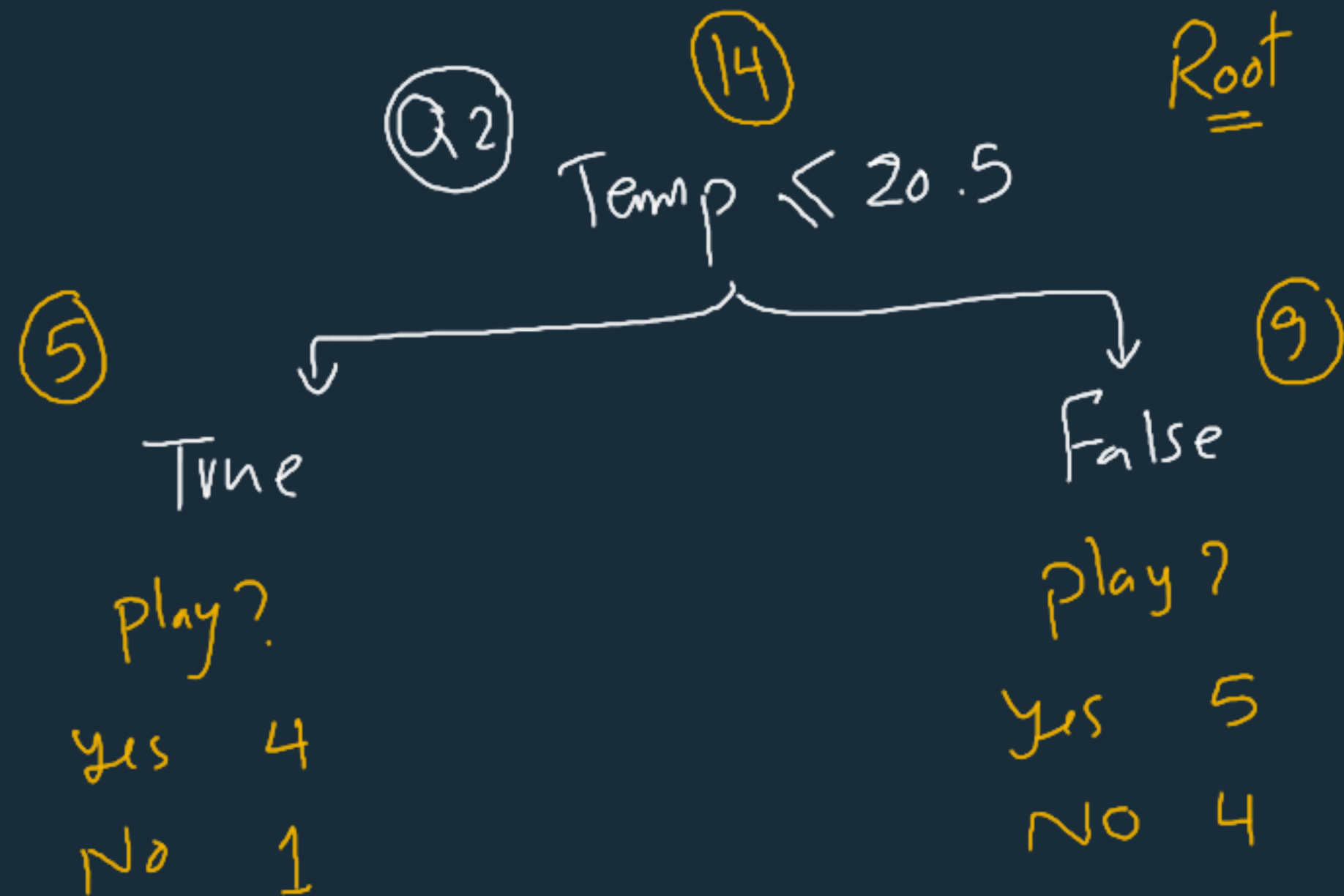
play?
yes 0
no 1

pure

False ⑬

play?
yes 9
no 4

$$\frac{1}{14} \left[\underbrace{1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2}_0 \right] + \frac{13}{14} \left[\underbrace{1 - \left(\frac{9}{13}\right)^2 - \left(\frac{4}{13}\right)^2}_{0.43} \right] = \underline{0.4}$$



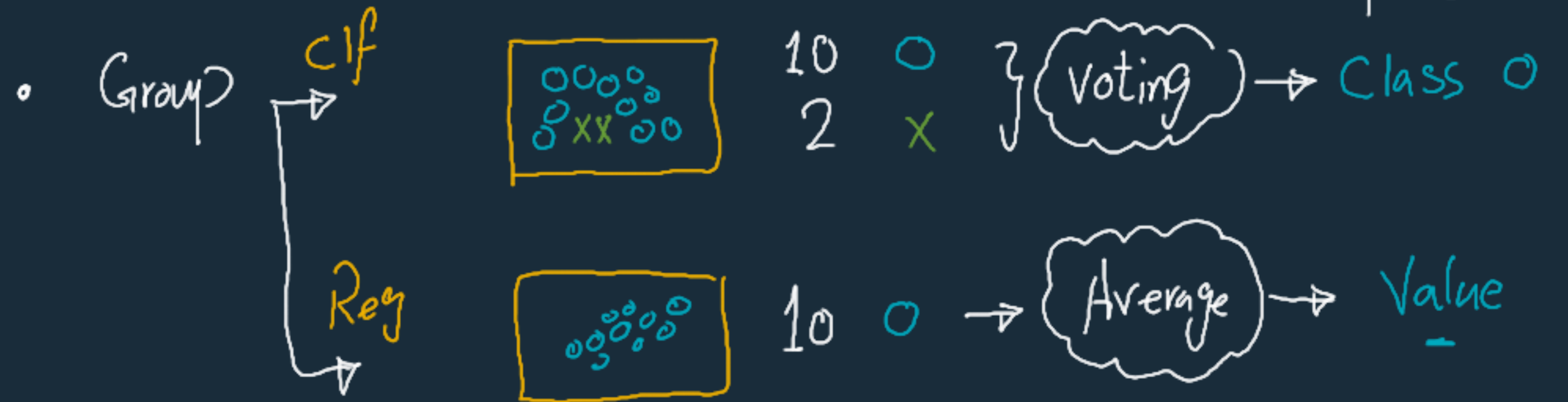
$$\frac{5}{14} \left[1 - \left(\frac{4}{5} \right)^2 - \left(\frac{1}{5} \right)^2 \right] + \frac{9}{14} \left[1 - \left(\frac{5}{9} \right)^2 - \left(\frac{4}{9} \right)^2 \right] = 0.43$$

$$G_{Q1} < G_{Q2}$$

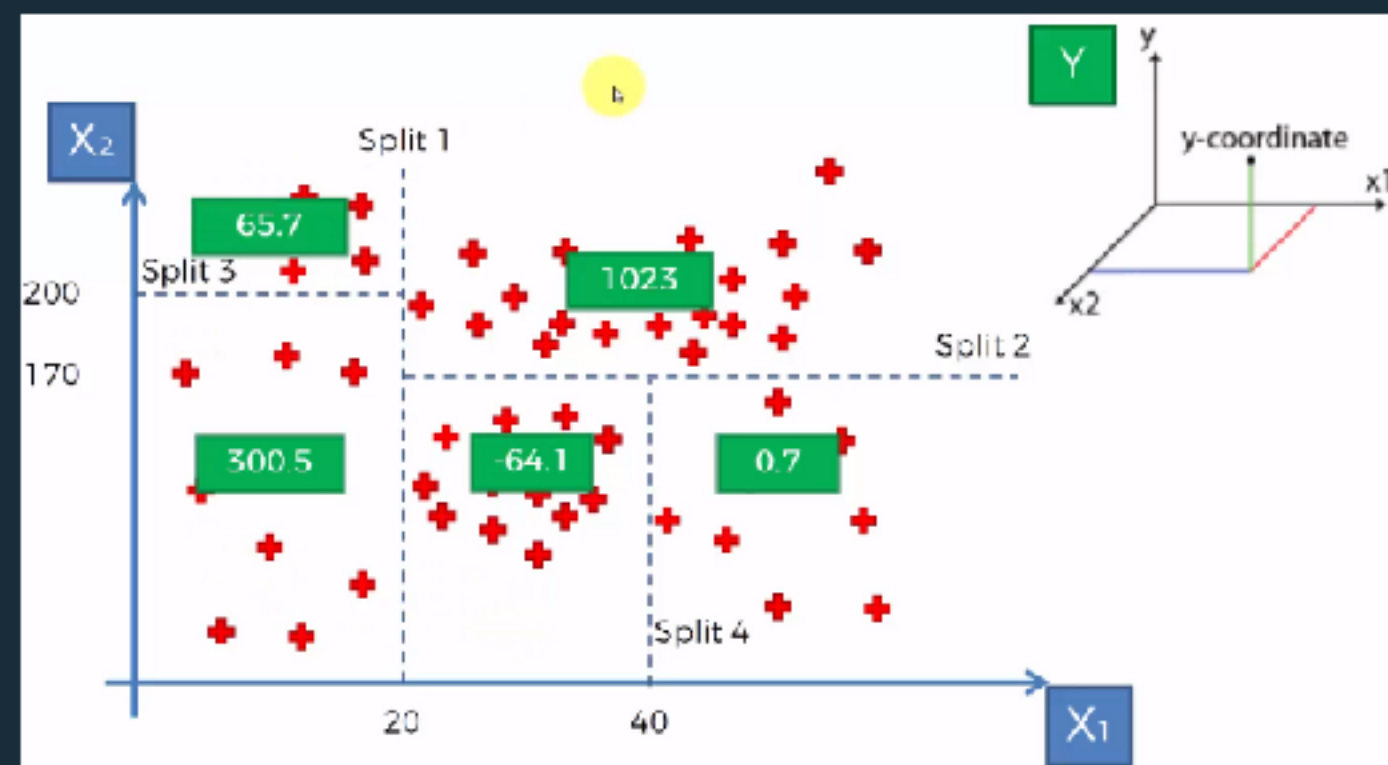
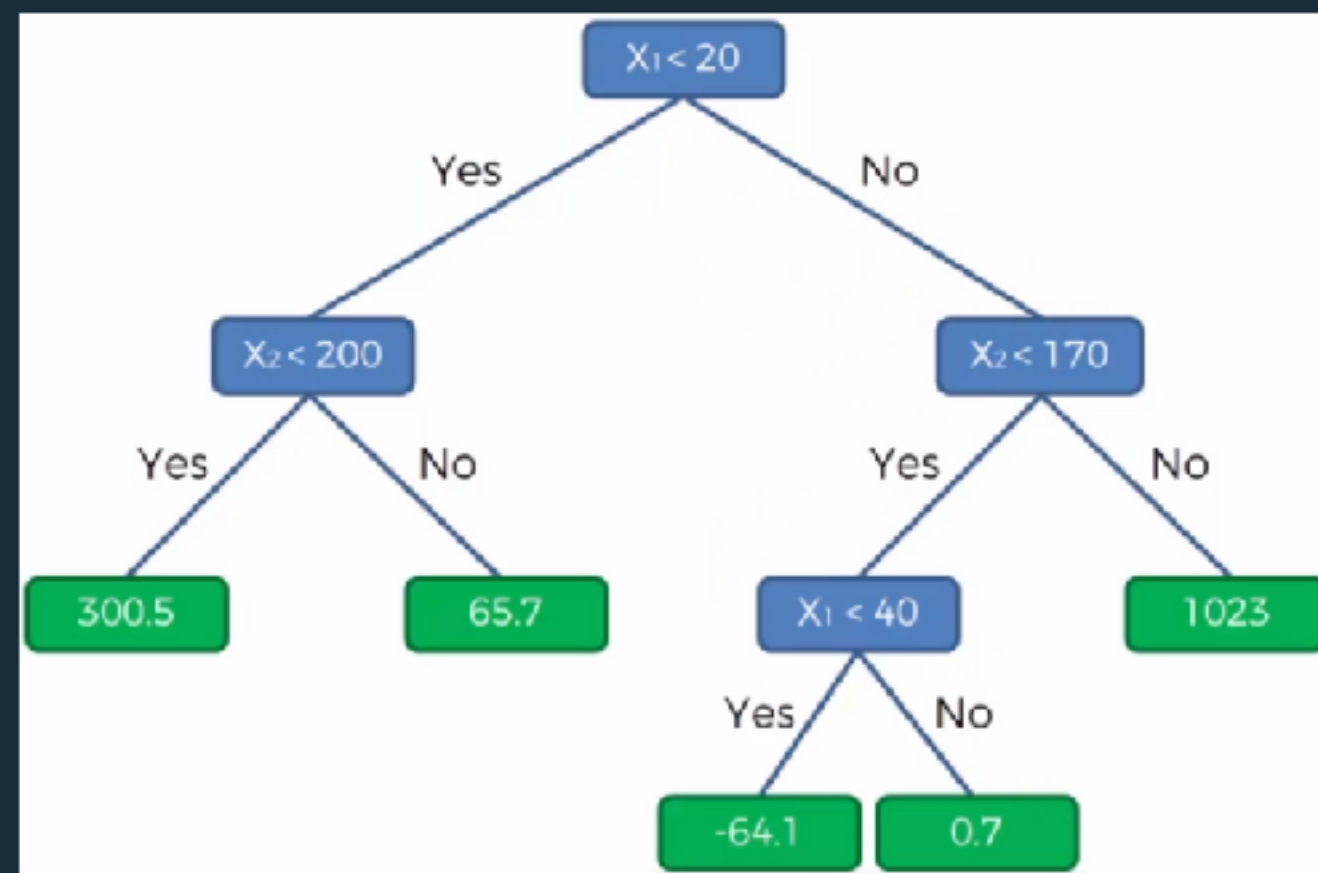
Decision Tree Regressor

- Question $\begin{cases} \text{True} \\ \text{False} \end{cases}$

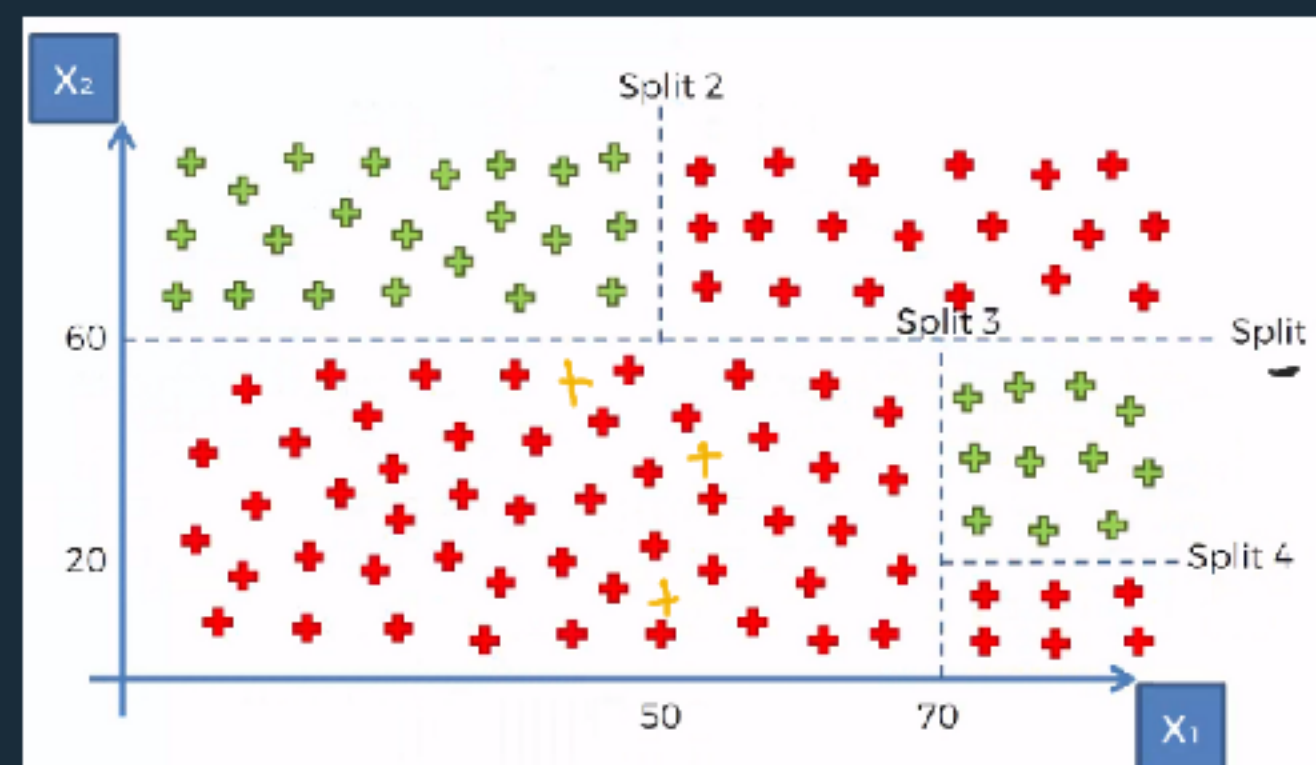
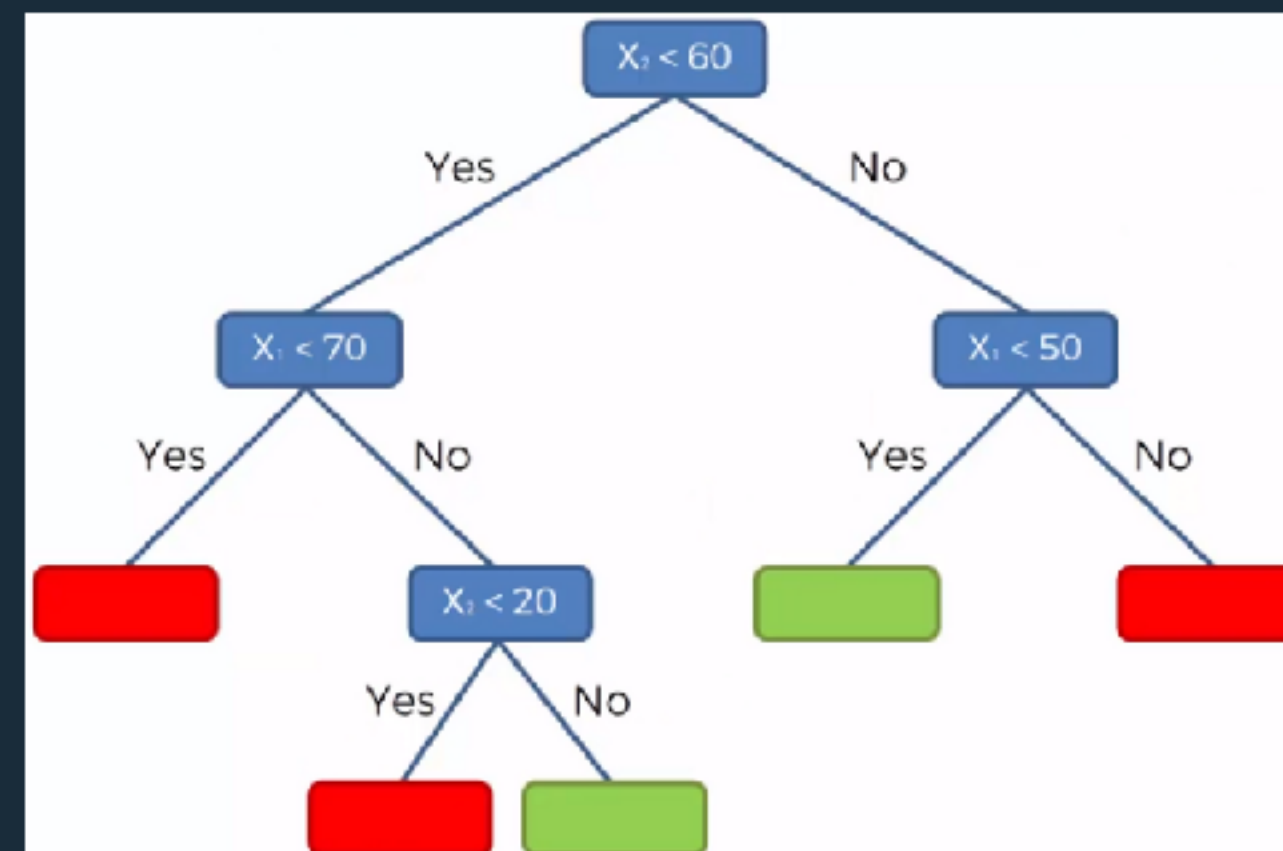
- Selection $\rightarrow$ Best Question $\begin{cases} \text{Attribute} \\ \text{Threshold} \end{cases}$



Regression



Classification



Equation 6-4. CART cost function for regression

$$J(k, t_k) = \frac{m_{\text{left}}}{m} \text{MSE}_{\text{left}} + \frac{m_{\text{right}}}{m} \text{MSE}_{\text{right}} \quad \text{where} \quad \begin{cases} \text{MSE}_{\text{node}} = \sum_{i \in \text{node}} (\hat{y}_{\text{node}} - y^{(i)})^2 \\ \hat{y}_{\text{node}} = \frac{1}{m_{\text{node}}} \sum_{i \in \text{node}} y^{(i)} \end{cases}$$

